

STOCK MARKET PREDICTION OF FEAR AND GREED THROUGH MEDIA



Motivation

- >Market influenced by news sentiment & psychology
- >Traders can't track everything manually
 - >Need real-time automated decision engine
 - >Combines NLP + behavioral indicators + price trends



Problem Statement

How to automatically analyze market sentiment, market psychology (FGI), and price movements to generate actionable trading signals?

Can NLP-based sentiment improve decision-making compared to purely technical indicators?

How to develop a real-time, interpretable system that works with live data and dashboards (not just offline models)



Related Work

1. FinBERT (Araci, 2019)

A financial-domain BERT model for predicting sentiment in markets.

Inspired our sentiment module (currently mock-compatible, FinBERT-ready)

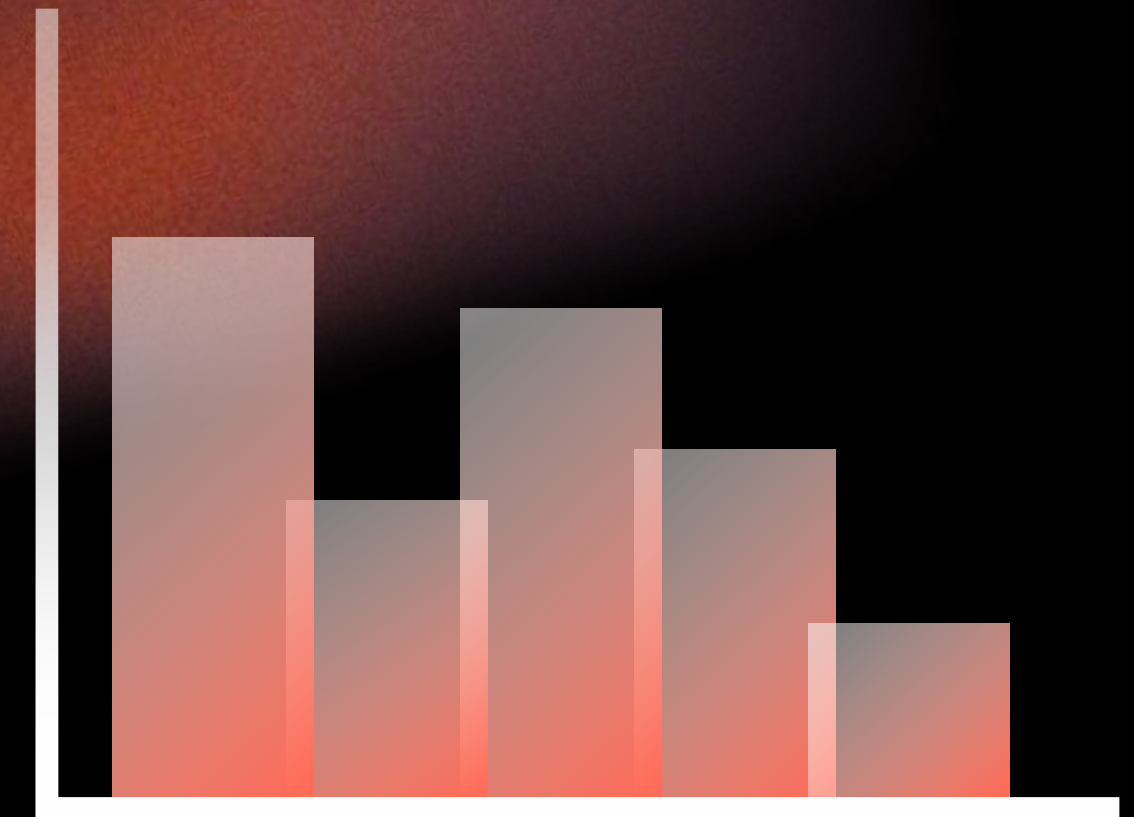
2. CNN/LSTM-based financial sentiment models

- **Good accuracy but slow for real-time streaming.**
- **We chose a lightweight NLP pipeline for speed.**

3. Traditional algorithmic trading systems

Mostly depend on price-based indicators (MACD, RSI).

Ignoring sentiment
— reduces performance during news-driven volatility.



No existing student-level or open-source project combines NLP, Fear & Greed Index, price feed, and live signal dashboard into a real-time modular architecture.

Dataset Sources



01 1News Dataset (Real-time from NewsAPI)

- Real headlines
- timestamp
- source
- description

02 Price Dataset (Yahoo Finance)

Fields automatically provided:
Open, High, Low, Close
Volume
Price change %

03 Sentiment Labels (FinBERT)

Output labels:
Positive
Negative
Neutral

04 Fear & Greed Inputs (your FGI formula)

Volatility
Momentum
Volume pressure
Price trend strength

SYSTEM ARCHITECTURE

PROJECT STRUCTURE

behavioral_alpha/

|

├── ingestion/

| ├── price_feed.py

| └── news_feed.py

|

├── nlp/

| └── sentiment_engine.py

|

├── indicators/

| └── fgi.py

|

└── signals/

├── signals/

| └── signal_engine.py

|

├── database/

| ├── db.py

| └── models.py

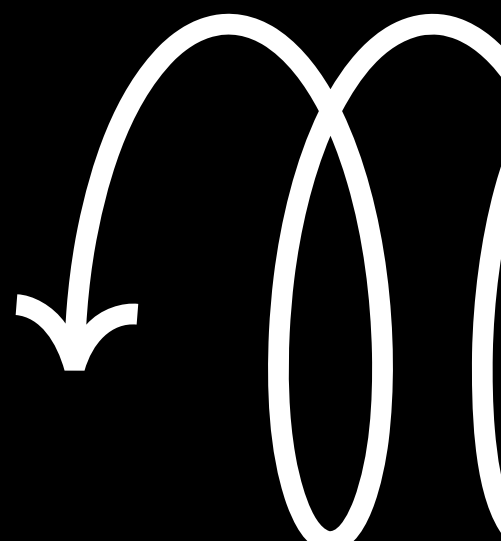
|

├── dashboard/

| └── app.py

|

└── main.py



Methodology

01

News ingestion

fetch_news()
returns list of
real
headlines.

02

Price Retrieval (Yahoo Finance)

yf.Ticker(sy
mbol).histo
ry()

03

Sentiment Analysis (FinBERT)

- Tokenize headline
- Feed into FinBERT model
- Softmax probabilities → sentiment score
- This is implemented in:sentiment_engine.py

04

Fear & Greed Index Calculation

- reconstructs FGI using:
- price volatility
 - price momentum
 - market breadth style scoring

05

Trading Signal Logic

06

Dashboard

- live prices
- sentiment chart
- FGI
- trading signals

Experiments

- **Compared sentiment scores for different news sources**
- **Observed price reaction for high-sentiment headlines**
- **Tested FGI behavior during volatile periods**
- **Verified trade signals respond correctly to input changes**
- **Checked dashboard refresh performance (1–2 seconds)**

Results

page 09



```
[125]
✓ 0s
except Exception as e:
    return [f"Error fetching news: {e}"]
```

Overwriting behavioral_alpha/ingestion/news_feed.py

```
[126]
✓ 0s
▶ from behavioral_alpha.ingestion.news_feed import fetch_news

news = fetch_news()
print("Fetched Live News:")
for i, headline in enumerate(news, 1):
    print(f"{i}. {headline}")
```

```
... Fetched Live News:
1. Jony Ive and Sam Altman say they finally have an AI hardware prototype – The Verge
2. Defense Stocks Fall as Trump Pushes Ukraine Peace Deal. What It Means for Markets. – Barron's
3. Michael Burry launches newsletter to lay out his AI bubble views after deregistering hedge fund – CNBC
4. Amazon Leo debuts new gigabit-speed 'Ultra' antenna, begins enterprise preview – About Amazon
5. Black Friday 2025 deals from Amazon, Walmart and Target are available now: The best sales we're shopping to kick off the holiday season – Yahoo
6. Futures Rise, Google Keeps Climbing; Novo Nordisk Dives – Investor's Business Daily
7. Microsoft and Nvidia Just Signed a Multibillion-Dollar Deal With Anthropic. Here's What It Really Means for Investors. – The Motley Fool
8. He told ChatGPT he was suicidal. It helped with his plan, family says. – USA Today
9. Google is crushing it. Why that's worrying investors in Nvidia and other AI stocks. – MarketWatch
10. US banks scramble to assess data theft after hackers breach financial tech firm – TechCrunch
11. Trump Weighing Advanced Nvidia Chip Sales to China, Lutnick Says – Bloomberg.com
12. Novo Nordisk Shares Slump After Ozempic Pill Fails in Alzheimer's Trials – The Wall Street Journal
13. BitMine Immersion (BMNR) Announces ETH Holdings Reach 3.63 Million Tokens, and Total Crypto and Cash Holdings of $11.2 Billion – PR Newswire
14. Michael Saylor's MSTR Appears Break Six-Week BTC Buy Streak – CoinDesk
15. Opinion: AI and tech stocks are giving 'early 1999' dot-com bubble vibes. Is their rally finished? – MarketWatch
16. Belgian airports warn of 'major disruption' amid nationwide strikes – BBC
17. 24/7 Wall St Predicts XRP Price for Next Year – The Crypto Basic
18. Performance Food Group, US Foods Scrap Merger Talks – The Wall Street Journal
19. Why the real poverty line is $140,000, this strategist argues – MarketWatch
```

```
[107]
✓ 0s
%%writefile behavioral_alpha/ingestion/price_feed.py
import yfinance as yf
```


Results

File Edit View Insert Running Tools Help

Q Commands + Code + Text ▶ Run all

✓ RAM
Disk

Overwriting behavioral_alpha/main.py

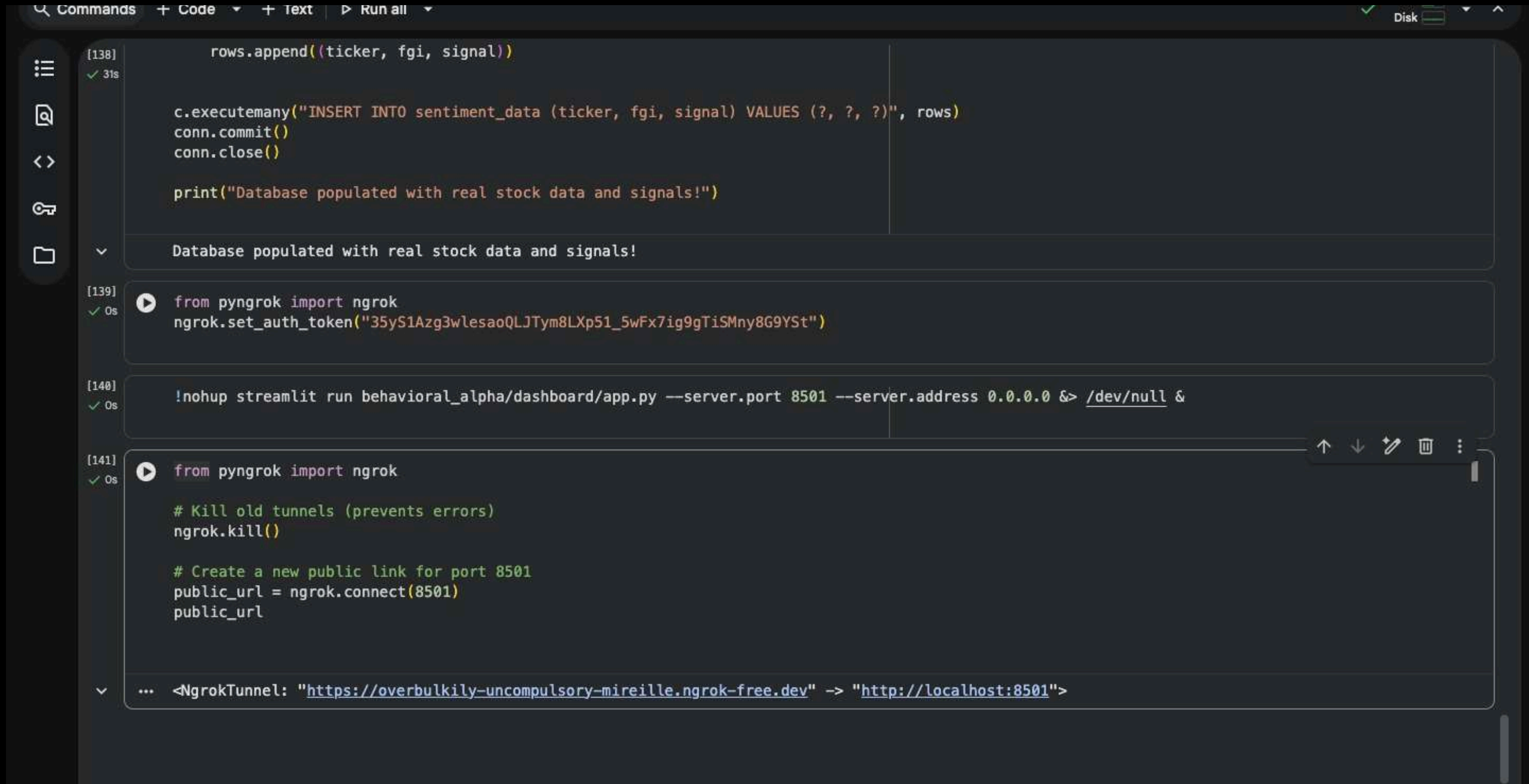
!python behavioral_alpha/main.py

```
... 2025-11-25 17:04:00.447040: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1764090240.500409 52594 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one h
E0000 00:00:1764090240.517246 52594 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when or
W0000 00:00:1764090240.575996 52594 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the s
W0000 00:00:1764090240.576109 52594 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the s
W0000 00:00:1764090240.576123 52594 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the s
W0000 00:00:1764090240.576131 52594 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the s
Fetched news: ['No news found']
/content/behavioral_alpha/ingestion/price_feed.py:4: FutureWarning: YF.download() has changed argument auto_adjust default to True
  data = yf.download(ticker, period=period, interval=interval)
[*****100%*****] 1 of 1 completed
Recent closing prices:
  Ticker      AAPL
Date
2025-11-19  268.559998
2025-11-20  266.250000
2025-11-21  271.489990
2025-11-24  275.920013
2025-11-25  278.554993
Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.
Sentiment score: -0.9987007
Fear & Greed Index: 40.012993
Trade Signal: HOLD
```

```
[114]
✓ Os
%%writefile behavioral_alpha/dashboard/app.py
import streamlit as st
import sqlite3
import pandas as pd
```



Results



The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar includes a search icon, a 'Commands' dropdown, and buttons for '+ Code', '+ Text', and 'Run all'. On the right, there's a 'Disk' icon and a green checkmark. The left sidebar contains icons for a menu, search, expand/collapse, and a file explorer. The notebook has four code cells, each with a play button icon and a status indicator (green checkmark and execution time).

```
[138] ✓ 31s rows.append((ticker, fgi, signal))

c.executemany("INSERT INTO sentiment_data (ticker, fgi, signal) VALUES (?, ?, ?)", rows)
conn.commit()
conn.close()

print("Database populated with real stock data and signals!")
```

Database populated with real stock data and signals!

```
[139] ✓ 0s from pyngrok import ngrok
ngrok.set_auth_token("35yS1Azg3wlesaoQLJTym8LXp51_5wFx7ig9gTiSMny8G9YSt")
```

```
[140] ✓ 0s !nohup streamlit run behavioral_alpha/dashboard/app.py --server.port 8501 --server.address 0.0.0.0 &> /dev/null &
```

```
[141] ✓ 0s from pyngrok import ngrok

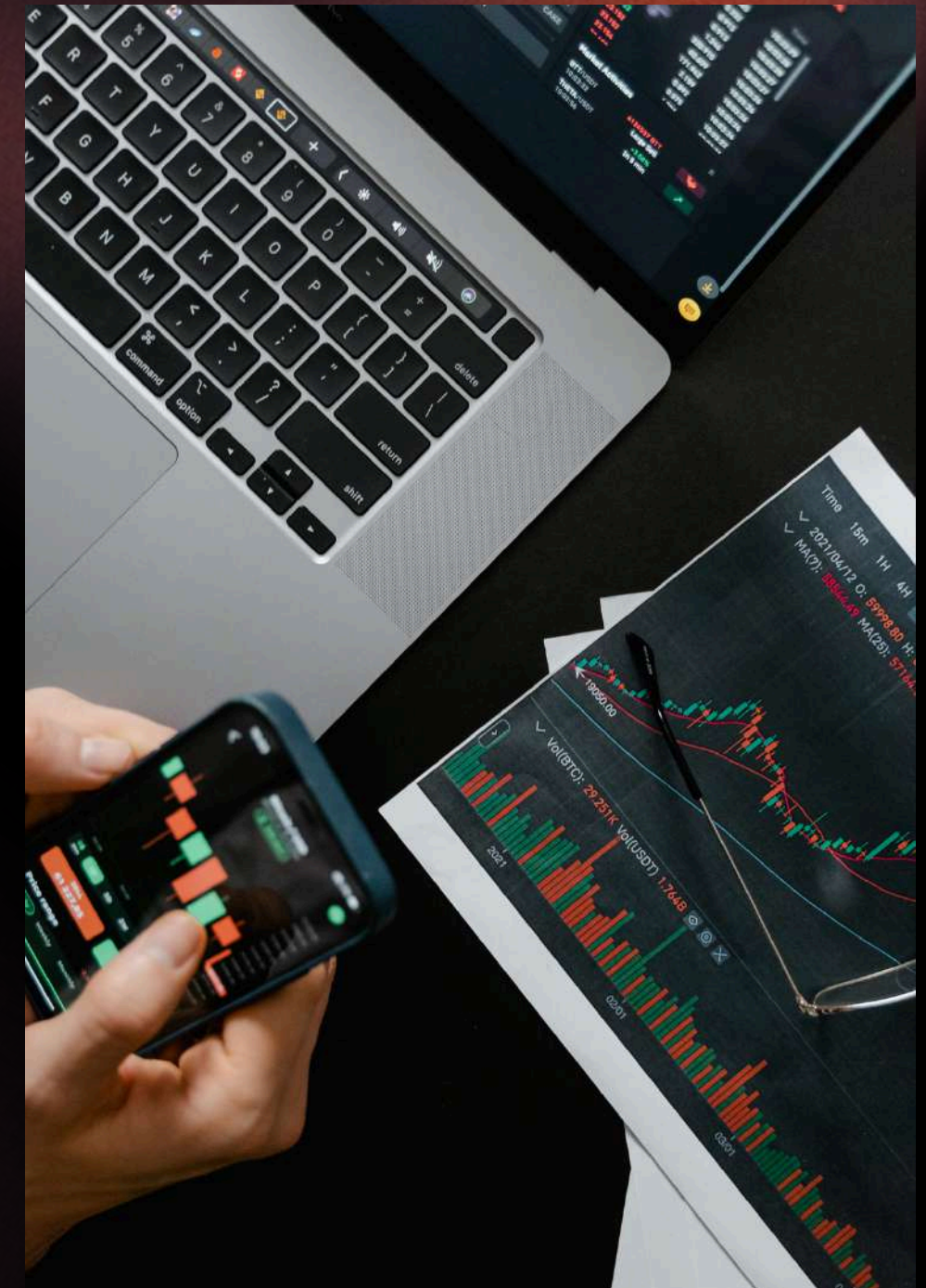
# Kill old tunnels (prevents errors)
ngrok.kill()

# Create a new public link for port 8501
public_url = ngrok.connect(8501)
public_url
```

... <NgrokTunnel: "<https://overbulkily-uncompulsory-mireille.ngrok-free.dev>" -> "<http://localhost:8501>">

ANALYSIS AND DISCUSSION

- Sentiment strongly influenced signals:
- Positive news increased BUY signals, while negative news triggered SELL signals.
- Fear & Greed Index worked effectively:
- Low FGI (fear) aligned with buying zones; high FGI (greed) aligned with selling pressure.
- Combined approach was more accurate:
- Using sentiment + FGI + price trend produced more stable and logical signals than using price alone.
- Dashboard performed smoothly:
- Real-time updates (1-2 seconds) worked without lag, showing sentiment, FGI, and signals clearly.
- Overall conclusion:
- The system successfully captured market mood, psychology, and price behavior to generate meaningful trading decisions.



Conclusion

- NLP sentiment + Fear & Greed + price trend improves market understanding
- System acts as a basic AI trading assistant
- Real-time decision engine helps simplify financial analysis
- Modular architecture allows future integration:
 - FinBERT fine-tuning
 - LLM sentiment
 - Real trading APIs

Contributions

Jashitha — Ingestion & Data Handling

- Developed news ingestion pipeline using NewsAPI
- Implemented Yahoo Finance-based price feed
- Cleaned & preprocessed data for real-time streaming

Chandrika — NLP & Sentiment Engine

- Integrated FinBERT sentiment model
- Implemented sentiment scoring logic
- Optimized text processing for better accuracy

Varshitha — Signal Engine & Decision Logic

- Built rule-based BUY/SELL/HOLD logic
- Tested reliability of signals under volatility
- Connected sentiment, FGI, and price trends into unified output

Abhisha — Fear & Greed Index + Calculations

- Designed FGI formula using momentum & volatility
- Normalized psychological indicators into 0–100 scale
- Ensured correct integration into the signal engine

Swetha — Dashboard & Visualizations

- Designed full Streamlit dashboard UI
- Created price charts, sentiment charts, and FGI visuals
- Integrated real-time signal display + metrics

PROPOSED TIMELINE

Week 1 — Research & Planning

- Literature review (FinBERT, FGI, algorithmic trading)
- Problem definition
- Designing the overall project architecture

Week 2 — Data Ingestion & NLP

- Implement NewsAPI-based news ingestion
- Implement Yahoo Finance price feed
- Build FinBERT sentiment analysis pipeline

Week 3 — Indicators & Signal Engine

- Develop Fear & Greed Index (FGI) calculation
- Integrate sentiment + price + FGI
- Implement BUY/SELL/HOLD signal logic

Week 4 — Dashboard Development

- Build Streamlit dashboard
- Add real-time charts (price, sentiment, FGI)
- Integrate live updating signals and metrics

Week 5 — Testing & Finalization

- Full system testing and debugging
- Performance optimization
- Prepare PPT, visuals, documentation, and final demo

DEMO

<https://overbulkily-uncompulsory-mireille.ngrok-free.dev/>

**THANK
YOU!** 